CBCS SCHEME

USN ☐☐☐☐☐☐☐☐☐☐ **BESCK104D**

First Semester B.E./B.Tech. Degree Examination, June/July 2025

Introduction to Mechanical Engineering

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	**M**	**L**	**C**
Q.1	**a.**	Explain briefly the emerging trends of mechanical engineering in Automobile and Energy Sectors.	**10**	**L2**	**CO1**
	b.	Discuss on the roles of mechanical engineers in different sectors.	**10**	**L2**	**CO1**
		OR			
Q.2	**a.**	Describe the construction and working of hydel power plant with a schematic diagram.	**10**	**L2**	**CO1**
	b.	Brief on the following environmental issues (causes and remedies) i) Global warming ii) Ozone layer depletion.	**10**	**L2**	**CO1**
		Module – 2			
Q.3	**a.**	Explain the working principal of drilling with schematic diagram.	**8**	**L2**	**CO2**
	b.	Illustrate the following operations of Milling with sketches. i) Plane milling ii) Slot milling.	**6**	**L3**	**CO2**
	c.	Explain the following operations of lathe i) Turning ii) Knurling.	**6**	**L2**	**CO2**
		OR			
Q.4	**a.**	Explain the different methods of manufacturing.	**4**	**L2**	**CO2**
	b.	Define Additive Manufacturing. Brief on steps involved in 3D printing with a block diagram.	**10**	**L2**	**CO2**
	c.	Describe the various components of CNC with schematic diagram.	**6**	**L2**	**CO2**
		Module – 3			
Q.5	**a.**	List the difference between Otto cycle and diesel cycle.	**4**	**L1**	**CO3**
	b.	Analyze the working of the 4 stroke diesel engine with sketch.	**10**	**L4**	**CO3**
	c.	Outline the classification of IC Engines.	**6**	**L1**	**CO3**
		OR			
Q.6	**a.**	Describe Electric Vehicles. Explain the components and working of Electric Vehicles.	**10**	**L2**	**CO3**
	b.	Describe Hybrid Vehicles. Explain the components and working of Hybrid Vehicles.	**10**	**L2**	**CO3**

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Module – 4					
Q.7	**a.**	Recite the classification of metals.	**5**	**L1**	**CO4**
	b.	Describe the three types of carbon steels with applications.	**9**	**L2**	**CO4**
	c.	Describe the following materials i) Graphite ii) Aluminum Alloy.	**6**	**L2**	**CO4**
OR					
Q.8	**a.**	Describe the construction and writing of gas welding process with neat sketch.	**10**	**L2**	**CO4**
	b.	List the differences between soldering brazing and welding.	**10**	**L1**	**CO4**
Module – 5					
Q.9	**a.**	Based on the configuration, explain the four types of robots with sketch.	**10**	**L3**	**CO5**
	b.	Define Mechatronics. List the difference between open loop and closed loop systems.	**10**	**L1**	**CO5**
OR					
Q.10	**a.**	Describe the communication models with respect to IoT.	**5**	**L1**	**CO5**
	b.	Define IoT. List the characteristics of IoT.	**5**	**L1**	**CO5**
	c.	Describe the basic elements of automation system with a block diagram.	**10**	**L2**	**CO5**

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